

DOWNLOAD_FIX

RuTracker Plugin Download Fix

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Problem

Downloads from RuTracker plugin were not starting in qBittorrent. The search worked and torrents could be sent to download, but they never actually started downloading. Direct upload of torrent files or magnet links worked fine.

Root Cause

The issue was in the `download_torrent` method in `plugins/rutracker.py`. The method was not properly: 1. Flushing the torrent file to disk before qBittorrent tried to read it 2. Syncing the file to ensure it's physically written to storage 3. Flushing stdout to ensure the output is immediately available to qBittorrent

Solution

Updated the `download_torrent` method to:

- 1. Explicitly flush and sync the file:**
 - Call `f.flush()` to flush Python's internal buffer
 - Call `os.fsync(f.fileno())` to ensure data is written to disk
- 2. Ensure file is closed before printing:**
 - Store the file path in a variable before closing
 - Print outside the `with` block to ensure file is fully closed
- 3. Flush stdout immediately:**
 - Call `sys.stdout.flush()` to ensure output is available to qBittorrent

Code Changes

Before:

```
def download_torrent(self, url: str) -> None:
    """Download torrent file and print filename + URL as required
    by API"""
    logger.info("Downloading {}...".format(url))
    data = self._open_url(url)
    with tempfile.NamedTemporaryFile(suffix=".torrent",
        delete=False) as f:
        f.write(data)
        print(f.name + " " + url)
```

After:

```
def download_torrent(self, url: str) -> None:
    """Download torrent file and print filename + URL as required
    by API"""
    logger.info("Downloading {}...".format(url))
    data = self._open_url(url)
    with tempfile.NamedTemporaryFile(suffix=".torrent",
        delete=False) as f:
        f.write(data)
        f.flush()
        os.fsync(f.fileno())
        temp_path = f.name
    print(temp_path + " " + url)
    sys.stdout.flush()
```

Testing

Created comprehensive automated tests to verify the fix:

Test Categories

- 1. Unit Tests** (tests/test_plugin_unit.py)
 - Test output format (filepath url)
 - Test file creation
 - Test file flushing and syncing
 - Test stdout flushing
 - Test search result building
- 2. Integration Tests** (tests/integration/)
 - Test real connection to RuTracker (requires credentials)
 - Test actual torrent downloads
 - Verify torrent file validity
- 3. End-to-End Tests** (tests/test_e2e_download.py)
 - Test complete workflow with qBittorrent API
 - Test search and download through Web UI
 - Verify torrent starts downloading

4. **Verification Script** (tests/verify_fix.py)
- Simple standalone test to verify the fix
 - Tests all critical aspects of the download process

Running Tests

```
# Quick verification
python3 tests/verify_fix.py

# Run unit tests
python3 tests/test_plugin_unit.py

# Run integration tests (requires RuTracker credentials)
python3 tests/integration/

# Run E2E tests (requires running container and credentials)
python3 tests/test_e2e_download.py --direct

# Run all tests via test suite
./run-all-tests.sh --suite python
```

Verification

All tests pass successfully:

```
=====
Testing download_torrent fix
=====

1. Testing output format...
  ✓ Format is correct: filepath url
  ✓ File path: /tmp/tmpixb8vzut.torrent
  ✓ URL: https://rutracker.org/forum/dl.php?t=12345

2. Testing file creation...
  ✓ Torrent file exists
  ✓ File content is correct

3. Testing file flushing...
  ✓ File flush was called
  ✓ os.fsync was called

4. Testing stdout flushing...
  ✓ stdout.flush was called

=====
ALL TESTS PASSED!
=====
```

Impact

This fix ensures that: - Downloads from RuTracker plugin now start reliably - Torrent files are fully written to disk before qBittorrent reads them - The output format is immediately available to qBittorrent - No data is lost due to buffering

Related Files

- `plugins/rutracker.py` - Fixed `download_torrent` method
- `tests/test_plugin_unit.py` - Unit tests
- `tests/integration/` - Integration tests
- `tests/test_e2e_download.py` - End-to-end tests
- `tests/verify_fix.py` - Quick verification script
- `tests/novaprinter.py` - Mock module for testing
- `./run-all-tests.sh` - Updated to include Python tests